

CH 1 Error & Approximations

Truncation Errors

Truncation errors occur when an infinite series or iterative process is approximated by stopping at a finite number of terms. These errors commonly arise in:

- **Taylor Series Approximations:** Higher-order terms are ignored.
- **Numerical Differentiation and Integration:** Only a finite number of steps are considered.
- **Finite Difference Methods:** Discretization of continuous functions.

Example:

Approximating e^x using the Taylor series:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

If we truncate after the second-order term, the truncation error is the sum of the ignored terms.

Rounding Errors

Rounding errors occur due to the finite precision of digital computers. They arise when:

- A number cannot be exactly represented in the machine's memory.
- Computations involve floating-point arithmetic, leading to small deviations.

Example: In a system that stores only four decimal places, representing $\pi=3.141592653$ as 3.1416 introduces a rounding error.

Fixed-Point Arithmetic

- Represents numbers with a fixed number of decimal places.
- Suitable for financial and monetary calculations where precision is crucial.
- Limited in range since it cannot represent very large or very small numbers efficiently.

Floating-Point Arithmetic

- Uses a scientific notation format: $N=M \times B^E$, where M is the mantissa, B is the base (usually 2 in binary systems), and E is the exponent.
- Allows representation of a much larger range of values.
- Used in scientific and engineering computations.
- **Example:** The number **0.0000345** can be written in floating-point as 3.45×10^{-5} .

Propagation of Errors

Error propagation refers to how errors accumulate and amplify through a sequence of computations. There are two main types:

1. **Additive Error Propagation:** If numbers with errors are added or subtracted, the total error is the sum of the absolute errors. **Example:**

If $x = 5.2 \pm 0.1$ and $y = 3.1 \pm 0.05$, then:

$$(x + y) = (5.2 + 3.1) \pm (0.1 + 0.05) = 8.3 \pm 0.15$$

- 2. Multiplicative Error Propagation:** When multiplying or dividing values, relative errors add up. **Example:**

If $x = 5.2 \pm 0.1$ and $y = 3.1 \pm 0.05$, then the relative error in multiplication is:

$$\frac{\Delta z}{z} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

Significant Figures

Significant figures determine the precision of a numerical value.

- The number of **reliably known digits** in a number.
- Includes all digits except **leading zeroes**.

Rules for Counting Significant Figures

1. **Non-zero digits** are always significant. Example: 235 has **3** significant figures.
2. **Leading zeros** are NOT significant. Example: 0.0045 has **2** significant figures.
3. **Trailing zeros** in a decimal number are significant. Example: 3.450 has **4** significant figures.
4. **Trailing zeros in a whole number without a decimal** are NOT significant. Example: 3400 has **2** significant figures (unless written as 3.400×10^3).
5. **Exact numbers** (like 100 people) have infinite significant figures.

Rounding Off Rules

- If the digit to be removed is **less than 5**, drop it.
- If the digit is **5 or greater**, round up the previous digit.

Example:

- 3.146 rounded to **3 significant figures** → **3.15**
- 0.04578 rounded to **2 significant figures** → **0.046**

Error Bounds

Error bounds define the range within which a computed value lies.

- The maximum possible deviation of an approximate value from the exact value.
- Helps in assessing the reliability of numerical computations.

2. Types of Error Bounds

1. **Absolute Error Bound:** The range within which the actual value lies. Example: If $x = 5.23 \pm 0.02$, then the absolute error bound is **0.02**.
2. **Relative Error Bound:** The absolute error divided by the true value. Example: If $x = 5.23 \pm 0.02x$, the relative error is: $0.02/5.23 = 0.0038$.
3. **Percentage Error Bound:** Relative error $\times 100$. Example: $0.0038 \times 100 = 0.38\%$.

3. Interval Representation If an approximate value is given as $x \pm \delta x$, the true value is in the range: $(x - \delta x) \leq \text{True Value} \leq (x + \delta x)$.